

Experiment 5: Induced EMF and working principal of a transformer

1. Objectives:

- (a) To verify the concept of induced emf
- (b) To calculate the turns-ratio of a transformer
- (c) To verify the effect of frequency on a transformer

2. Background:

The transformer consists of two coils: the primary coil (input) and the secondary coil (output). These coils are wound around a core made of a ferromagnetic material (typically iron), which serves to efficiently channel the magnetic flux between the coils, thereby enhancing the process of mutual induction.

A change in current passing through the primary coil of a transformer induces an electromotive force (EMF) within the same coil. This phenomenon is known as self-induction, and it is governed by Faraday's Law of Electromagnetic Induction.

$$\mathcal{E}_L = -L \frac{di}{dt}$$

Here, $\mathcal{E}_L$ is the self-induced EMF, L is the inductance of the coil, and $\frac{di}{dt}$ is the rate of change of current. The direction of the induced EMF opposes the change in magnetic flux that caused it, and can be explained according to Lenz's Law.

A transformer is a widely used device that works on the principle of mutual induction. It consists of two coils with different number of turns wound around an iron core. The primary winding (called primary) of N_p turns is connected to an alternating current generator whose emf at any time t is given by

$$\mathcal{E}(t) = \mathcal{E}_m \sin \omega t$$

$\mathcal{E}(t)$ refers to the instantaneous EMF at time t , $\mathcal{E}_m$ is the maximum (peak) value of the EMF.

This alternating voltage drives an alternating current ($I(t) = \frac{\mathcal{E}(t)}{R} = \frac{\mathcal{E}_m \sin \omega t}{R} = I_m \sin \omega t$) through the primary coil. The sinusoidally changing primary current produces a sinusoidally changing magnetic flux in the iron core. The core acts to strengthen the flux and to bring it through the secondary winding (called secondary). As the flux varies, it induces an emf in each turn of the secondary. So, as the magnetic flux links with the secondary core via core, the changing magnetic field induces an EMF in the secondary coil according to Faraday's law of Electromagnetic induction.

$$\mathcal{E}_{\text{ind}} = -N \frac{d\phi}{dt}$$

Here, N is the number of turns in the coil, $\frac{d\phi}{dt}$ is the rate of change of magnetic flux. Here, this phenomenon where current in the primary coil induces a voltage in the secondary coil is known as **mutual induction**.

The induced EMF per turn $\mathcal{E}_{\text{turn}}$ is the same in the primary and the secondary. Across the primary, the voltage V_p is the product of $\mathcal{E}_{\text{turn}}$ and the number of turns in the primary, N_p , i.e., $V_p = \mathcal{E}_{\text{turn}} * N_p$

Similarly, for the secondary side, $V_s = \mathcal{E}_{\text{turn}} * N_s$

Thus, we can write, $\varepsilon_{turn} = \frac{V_p}{N_p} = \frac{V_s}{N_s}$

If $N_s > N_p$, the device is called a step-up transformer because the secondary voltage is greater than the primary voltage.

If $N_p > N_s$, the device is called a step-down transformer because the secondary voltage is smaller than the primary voltage. However, for a transformer, the ratio of the voltages of across two terminals- called the turns ratio - is equal to the ratio of the number of turns of the corresponding terminals.

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

In this experiment, a step-down center-tapped transformer (12V – 0 – 12V) will be used.

3. Procedures and Observations:

Part 1: Induced EMF:

1. Complete the circuit diagram, as shown in Figure 1.
2. Connect the primary of the transformer to a DC power supply. Set the voltage to 5V. Adjust the current knob, if necessary.
3. Connect the secondary of the transformer to one of the channels of an oscilloscope.
4. Make sure the channel is set to DC coupled with volt/div set at **0.2V**.
5. Measure the voltage. It will be zero volt.
6. Observe the voltage on the oscilloscope closely as you turn on and off the power supply. (**Be patient; wait for at least 30 seconds between turning on and off**)

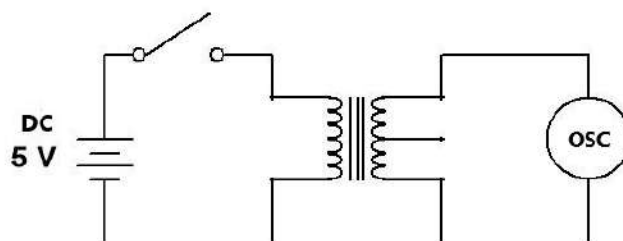


Figure 1

7. What did you notice? Note your observation below.

8. Using the oscilloscope, measure the amplitude of the induced voltage as you turn on and turn off the power supply. Use the STOP button to measure the amplitude of the voltage.

Turn on: Amplitude of the induced voltage _____

Turn off: Amplitude of the induced voltage _____

Part 2: Calculating the Turns Ratio

1. Complete the circuit diagram, as shown in Figure 2.
2. Connect the function generator to the primary and oscilloscope to both primary and secondary terminals of the transformer.
3. Set the function generator to produce **sinusoidal signal** with a frequency of 50Hz.
4. Use the function generator to set the input voltage and measure the peak-to-peak output voltage from the oscilloscope.
5. Calculate the turns-ratio of the transformer to complete Table 1.
6. Connect the equipment according to the circuit diagram in Figure 3 and repeat the steps above to fill out Table 2.

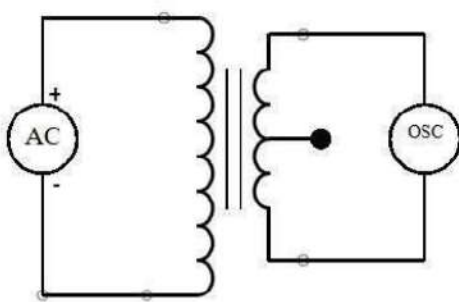


Figure 2

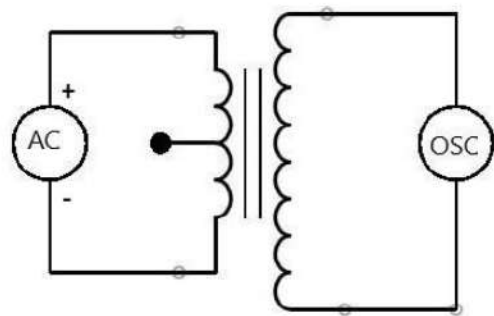


Figure 3

Lab Report:

Date:	
Name of the Students and IDs	(1)
	(2)
	(3)

Data Tables:

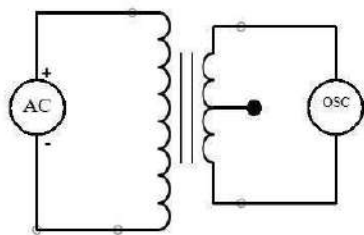


Table 1 (Use the set up shown in above figure)

Primary Voltage ($V_{P(p-p)}$)	Secondary Voltage ($V_{S(s-s)}$)	$\frac{V_{P(p-p)}}{V_{S(s-s)}}$
1		
2		
3		
4		
5		
6		
7		
8		

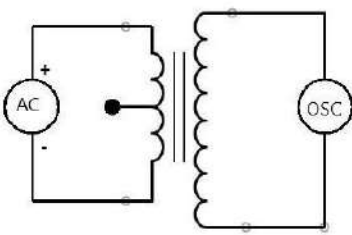


Table 2 (Use the set up shown in above figure)

Secondary Voltage ($V_{S(s-s)}$)	Primary Voltage ($V_{P(p-p)}$)	$\frac{V_{P(p-p)}}{V_{S(s-s)}}$
1		
2		
3		
4		
5		
6		
7		
8		

Task and Questions:

#1: Explain briefly your observation in Part #1 of the experiment with reference to the induced EMF.

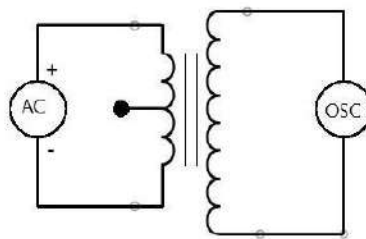
#2: What happens to the amplitude of the induced voltage if the number of turns in the primary coil (output terminal) of the transformer is reduced?

#3: Does a transformer work when a DC current flows through the primary? If not, why?

#4: Use the data in Table 1 to calculate the average turns-ratio of the transformer.

#5: If the transformer in this experiment has 400 turns in the primary, how many turns does it have in its secondary? Use the data from question #4.

#6: If the frequency of the input signal is increased to 250 kHz, what happens to the output? You can build the following circuit to answer to this question. [Give input voltage 5V (peak-peak) and observe the output]



Results:

Discussion: